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A Comparative Performance Analysis of Structural Equation Modeling Approaches: A Monte Carlo Simulation Study

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Abstract:

This Monte Carlo simulation study compared the performance of three structural equation modeling approaches CB-SEM, PLS-SEM, and GSCA within a mediation model under varying sample sizes, data distributions, and indicator counts. Using full factorial design with 500 replications per scenario, results revealed that CB-SEM provided the most accurate estimates under normality and larger samples ($N \geq 250$), yet suffered convergence issues in small or severely non-normal conditions. PLS-SEM demonstrated superior convergence rates and statistical power across all scenarios, albeit with systematic attenuation bias. GSCA offered balanced intermediate performance, particularly robust under distributional violations. Increasing indicators per construct benefited CB-SEM most substantially. No single method proved universally superior; rather, selection should align with research objectives, sample availability, and data characteristics. These evidence-based findings offer practical guidance for biostatisticians and health researchers in choosing appropriate SEM methodologies.

Keywords: CB-SEM, PLS-SEM, GSCA, Monte Carlo Simulation, Mediation Analysis.

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1 Introduction

Structural Equation Modeling (SEM) provides a comprehensive framework for simultaneously examining complex relationships among latent and observed variables [JÖRESKOG \(1973\)](#) and has become integral to behavioral, social, and increasingly biomedical research [MACCALLUM \(2000\)](#). In the health sciences, its relevance is underscored by the latent nature of key constructs—such as health-related quality of life, functional status, and treatment adherence—which are measurable only through multiple indicators. Three principal estimation approaches have emerged over the past five decades. Covariance-Based SEM (CB-SEM) aims to reproduce the observed covariance matrix by minimizing the discrepancy between sample and model-implied matrices, typically employing maximum likelihood estimation under multivariate normality assumptions ([BOLLEN \(1989\)](#)). Partial Least Squares SEM (PLS-SEM), a variance-based approach, maximizes the explained variance in dependent latent variables through iterative ordinary least squares regression, requires no stringent distributional assumptions, and accommodates small samples (Wold, 1982, Hair et al., 2019). Generalized Structured Component Analysis (GSCA), introduced by Hwang and Takane (2004), employs an alternating least squares algorithm for simultaneous estimation of measurement and structural models, offering an integrated framework for both reflective and formative constructs [HWANG \(2004\)](#). Despite extensive comparisons between CB-SEM and PLS-SEM [REINARTZ \(2009\)](#); [CIVELEK \(2018\)](#), comprehensive evaluation of all three methods within a unified framework—particularly in mediation models—remains limited. Moreover, the role of indicator quantity per latent variable has received insufficient attention in simulation studies, despite classical measurement theory suggesting its impact on reliability and bias reduction [MARSH \(1998\)](#). This study addresses these gaps through a comprehensive Monte Carlo simulation comparing CB-SEM, PLS-SEM (with PLSc bias correction), and GSCA within a three-latent-variable mediation model, examining estimation accuracy, convergence rates, indicator quantity effects, and statistical power for detecting indirect effects across varying distributional conditions.

2 Materials and Methods

A Monte Carlo simulation with a full factorial design was conducted, manipulating four factors. The estimation method factor comprised CB-SEM using the lavaan package (v0.6-18) [ROSSEEL \(2012\)](#) with ML estimation (switching to robust MLR for non-normal data); PLS-SEM using the cSEM package (v0.6.1) [RADEMAKER \(2025\)](#) with the consistent PLSc algorithm to correct attenuation bias [DIJKSTRA \(2015\)](#); and GSCA using the gesca package (v1.0) [KIM \(2017\)](#) with alternating least squares (maximum 200 iterations, convergence threshold 10^{-5}). Sample size included five levels ($N = 50, 100, 250, 500, 1000$), and

data distribution comprised four conditions: multivariate normal, skewed normal (skewness 1.5, kurtosis 3.0), $t(5)$ (heavy tails), and gamma (shape 2.0, severely asymmetric). The number of indicators per latent variable was set at 3, 5, or 7. The full crossing of these factors produced 180 unique scenarios, each replicated 500 times, yielding 90,000 simulated datasets. The theoretical model was a simple mediation with three latent variables—-independent (ξ), mediator (η_1), dependent (η_2)—specified by the structural equations $\eta_1 = \gamma_1\xi + \zeta_1$ and $\eta_2 = \beta_1\eta_1 + \gamma_2\xi + \zeta_2$. True population parameters were $\gamma_1 = 0.6$, $\beta_1 = 0.5$, and $\gamma_2 = 0.3$, giving an indirect effect of 0.3 and a total effect of 0.6. Structural errors were drawn from a normal distribution with variance scaled to achieve intended R^2 values. All factor loadings were fixed at 0.7, yielding composite reliabilities between 0.75 and 0.85 depending on indicator count, consistent with recommendations for basic research ?; indicator error variances were $1 - \lambda^2$. Data generation proceeded by first creating factor scores using the Fleishman (1978) power transformation for non-normal distributions [FLEISHMAN \(1978\)](#), applying the Vale and Maurelli (1983) procedure for skewed conditions [VALE \(1983\)](#), then computing endogenous variables and generating observed indicators via $X = \Lambda F + \varepsilon$. For each successfully converged replication, estimated path coefficients (γ_1 , β_1 , γ_2), the indirect effect ($\gamma_1 \times \beta_1$), standard errors, 95% confidence intervals, p -values, and, for CB-SEM, fit indices (CFI, RMSEA, SRMR) were extracted. Performance was assessed through relative bias, RMSE, convergence rate, and statistical power for the indirect effect at $\alpha = 0.05$, with acceptable bias defined as $\pm 10\%$ [HOOGLAND \(1989\)](#).

3 Results

Convergence Rates (Table 1): CB-SEM showed convergence problems at $N=50$, especially under non-normality (e.g., 42% for gamma), while PLS-SEM exceeded 98% across all conditions and GSCA remained stable (minimum 87% at $N=50$ gamma).

Relative Bias: CB SEM had negligible bias ($< 2\%$) under normality with $N \geq 250$, but up to -18% at $N = 50$ under non normality. PLS SEM consistently showed attenuation bias ($\approx -25\%$ at $N = 50$, $-5-7\%$ at $N = 1000$), aligning with known PLS attenuation [DIJKSTRA \(2015\)](#). GSCA displayed intermediate bias ($3-8\%$ under normality) and outperformed CB SEM under severe non normality.

RMSE of Indirect Effect (Table 2): Increasing indicators from 3 to 7 reduced RMSE by 35–50%, with CB SEM benefiting most. At $N=100$, PLS SEM had the lowest RMSE; at $N=500$ with 7 indicators, CB SEM achieved the best precision (RMSE=0.023).

Table 1: Convergence Rates (%) for 3 Indicators

Method	Sample Size	Normal	Skewed	$t(5)$	Gamma
CB-SEM	50	76.2	68.4	59.6	42.0
	100	94.8	91.2	88.0	79.4
	250	99.6	98.8	97.4	94.2
	≥ 500	100	99.8	99.6	98.8
PLS-SEM	50	99.2	98.8	98.4	97.6
	≥ 100	100	100	99.8	99.6
GSCA	50	96.4	94.2	91.8	87.2
	≥ 100	99.8	99.4	99.0	97.8

Table 2: RMSE (True indirect effect = 0.3, normal distribution)

Indicators	Method	$N = 100$	$N = 250$	$N = 500$
3	CB-SEM	0.092	0.054	0.038
	PLS-SEM	0.078	0.052	0.042
	GSCA	0.085	0.056	0.040
5	CB-SEM	0.071	0.041	0.029
	PLS-SEM	0.065	0.044	0.036
	GSCA	0.073	0.043	0.031
7	CB-SEM	0.058	0.033	0.023
	PLS-SEM	0.056	0.038	0.031
	GSCA	0.060	0.035	0.025

Statistical Power: CB SEM required $N=200$ to reach 80% power; PLS SEM achieved 0.48 at $N=50$, 0.71 at $N=100$, and 80% power at $N=140$. GSCA had intermediate power (0.63 at $N=100$, 0.85 at $N=250$), confirming PLS SEM's efficiency under sample size constraints (Reinartz et al., 2009). **Effects of Non Normality (Table 3):** Under gamma distribution ($N=250$, 5 indicators), standard CB SEM (ML) showed -11.5% bias, exceeding the $\pm 10\%$ threshold; robust MLR reduced this to -7.2% . PLS SEM (PLSc) and GSCA displayed stable bias across distributions, with PLS SEM yielding the lowest bias (-4.1%) under gamma, highlighting its robustness for skewed data.

Table 3: Relative Bias (%) in Estimation of $\beta_1 = 0.5$ at $N = 250$ with 5 Indicators

Method	Normal	Skewed	$t(5)$	Gamma
CB-SEM (ML)	-1.2	-4.8	-6.3	-11.5
CB-SEM (MLR)	-1.3	-2.9	-3.7	-7.2
PLS-SEM (PLSc)	-5.6	-5.2	-4.9	-4.1
GSCA	-3.1	-3.4	-3.8	-4.5

Sensitivity to Indicator Quantity: In CB SEM, adding indicators enriches the covariance matrix, directly lowering standard errors and RMSE. In PLS SEM and GSCA, improvement mainly occurs through increased composite reliability and attenuation reduction, but at a slower rate. With 7 indicators, CB SEM achieved a $> 50\%$ RMSE reduction versus the 3 indicator case, compared to 35% for PLS SEM, demonstrating the covariance based method's greater ability to exploit rich measurement models.

4 Discussion

4.1 Interpretation of Key Findings.

This first comprehensive three-method SEM simulation confirms that no single approach dominates all conditions. CB-SEM provides the most accurate, minimally biased estimates under ideal conditions (normality, $N \geq 250$, ≥ 5 indicators), consistent with REINARTZ (2009). However, performance sharply declines in small samples ($N < 100$), with non-convergence exceeding 40%, and under severe non-normality (gamma bias -11.5%), challenging claims of its robustness REINARTZ (2009). The robust MLR estimator substantially mitigates this bias. PLS-SEM's strengths are near-perfect convergence even at $N = 50$ and superior statistical power, making it ideal for exploratory or sample-constrained settings HAIR (2019), though it consistently exhibits attenuation bias due to reliance on linear composites rather than covariance reproduction DIJKSTRA (2015). PLSc correction reduces but does not eliminate this bias, aligning with CIVELEK (2018). GSCA offers a balanced intermediate performance—outperforming standard CB-SEM under non-normality and converging more often than CB-SEM in small samples—but remains limited by lower researcher familiarity and software support.

4.2 Importance of Indicator Quantity.

A novel contribution is documenting that CB-SEM benefits substantially more from additional indicators ($\approx 50\%$ RMSE reduction from 3 to 7 indicators) than does PLS-SEM ($\approx 35\%$), because CB-SEM fully exploits

the enriched covariance matrix, whereas component-based methods disregard some of that information. Practically, CB-SEM users should aim for at least 5–7 indicators, while for PLS-SEM, increasing beyond 5 yields diminishing returns.

4.3 Limitations and Future Directions.

The study used a simple three-variable mediation model; performance may differ in more complex structures (moderators, multiple mediators, longitudinal designs). Factor loadings were fixed at 0.7, whereas real-world loadings vary. Complete data were assumed, yet missing data are common and handled differently across methods. Bayesian SEM, despite its advantages for small samples, was not examined. Furthermore, *gesca*'s bootstrap options are less mature than those of *lavaan* or *cSEM*. Future work should extend simulations to more complex models, systematically vary loading quality, evaluate performance under MCAR, MAR, and MNAR mechanisms, and compare Bayesian with frequentist approaches.

4.4 Practical Recommendations.

Based on the study findings, the following decision-making guide (Table 4) is proposed for selecting SEM methods:

Table 4: Decision Guide for SEM Method Selection

Research Condition	Recommended Method	Considerations
$N \geq 250$, approx. normal distribution, theory testing	CB-SEM (ML)	Gold standard; report fit indices
$N \geq 250$, moderate non-normality	CB-SEM (MLR)	Use robust Huber-White SEs
$N < 100$, any distribution	PLS-SEM (PLSc)	Interpret as exploratory; account for attenuation bias
$N = 100\text{--}250$, severe non-normality	GSCA or PLS-SEM	GSCA balances precision and stability
3 or fewer indicators	CB-SEM (with caution)	Just-identified model; all methods conservative
7 or more indicators	CB-SEM	Maximise covariance information
Prediction or theory development	PLS-SEM	Focus on R^2 and predictive relevance, not model fit

Supplementary note: Where possible, run sensitivity analyses with multiple methods; convergence of CB-SEM and PLS-SEM results increases confidence.

5 Conclusion

This Monte Carlo simulation provides evidence-based insights into the performance of CB-SEM, PLS-SEM, and GSCA within a mediation model. Four key findings emerge: (1) a precision–stability trade-off exists—CB-SEM excels under ideal conditions but struggles in small or severely non-normal samples, PLS-SEM ensures convergence at the cost of attenuation bias, and GSCA occupies a middle ground; (2) PLS-SEM demonstrates superior statistical power, making it ideal for exploratory and sample-constrained research; (3) measurement design is critical, with CB-SEM deriving disproportionate benefit from additional indicators; and (4) no method is universally superior, requiring context-dependent selection based on research objectives, resources, and data characteristics. For biostatisticians and health researchers, the study offers a practical, evidence-based framework for informed SEM method selection, ultimately contributing to stronger evidence quality in biomedical research.

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